

Module 6 : Data Management

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Data Management: It is used to import/export bulk data to/from salesforce.

I want to insert 50 candidate records at a time.

Importing the data --> taking the data to salesforce org from some other application like excel, database etc.

Exporting the data --> taking the data from salesforce to any other external tool like excel, database etc.

Importing the data consists of :

1. Insert
2. Update
3. Upsert

Export consists of:

1. Export
2. Scheduled export

We have below 2 tools for import/Export:

1. Data Import Wizard: is an internal tool. Means it is available inside the salesforce org itself.
2. Dataloader.io: is an external tool. It is a cloud-based tool.

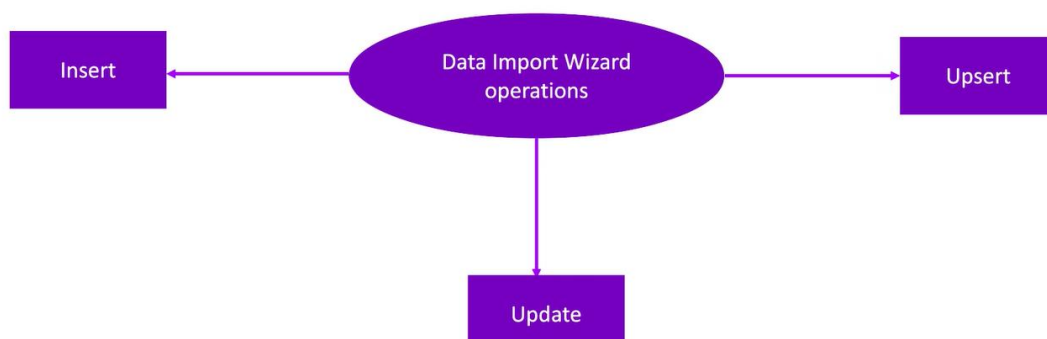
Data Management

Data management in Salesforce specifically refers to the processes involved in transferring bulk data into and out of a Salesforce system, while ensuring accuracy, consistency, and security.

Import Data

Understanding Data Import Wizard

1. **Data Import Wizard:** The Data Import Wizard is a tool within Salesforce that allows users to import(insert, update and upsert) data from CSV files directly into standard and custom objects.



Data Import Wizard

Data Import Wizard facilitates below import operations.

- 1. Insert:** The insert operation allows users to add new records to Salesforce. Users can import data from external sources (e.g., CSV files) into Salesforce objects such as Leads, Contacts, Accounts, Campaign members, Solutions and Custom objects.
- 2. Update:** The update operation enables users to modify existing records in Salesforce. Users can import data from external sources containing updated information for specific records, and Data Import Wizard updates the corresponding records in Salesforce based on field mappings.
- 3. Upsert (Update or Insert):** The upsert operation combines the functionalities of both insert and update operations. It allows users to insert new records into Salesforce and update existing records simultaneously based on field mappings.

Convert to csv file only then we can use it.

Common Errors encountered during Data Import using Data Import Wizard

Below are the common types of errors that can be encountered while importing data using Data Import Wizard.

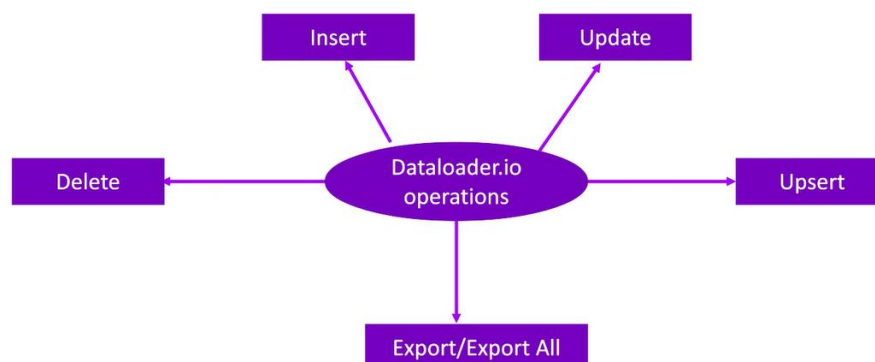
- 1. Required Field Errors :** Records have missing fields that are required by Salesforce.
- 2. Data Format Errors :** Source data is not in a valid format. For Example, Date and Number fields format.
- 3. Duplicate Record Errors:** Records being imported already exist in Salesforce and are detected as duplicates.
- 4. Field Mapping Errors:** Source fields are not mapped to Salesforce fields.

5. **Data Validation Rule Errors:** Records do not meet the criteria set by validation rules in Salesforce.
6. **Data Type Mismatch Errors:** The data type of the source field does not match the Salesforce field.
7. **Picklist Value Errors:** Source data contains values not present in the Salesforce picklist.
8. **Permission Errors:** User does not have the necessary permissions to import data or modify certain fields.

Using Dataloader.io to Insert Data

Understanding Dataloader.io

Dataloader.io is a cloud-based data management tool provided by Salesforce for importing, exporting, updating, and deleting large volumes of data. Dataloader.io also supports **Scheduled Import/Export**.



Insert System Fields

- We can use Dataloader.io to insert data into the Salesforce Org just like Data Import Wizard.
- With Dataloader.io, we can import the **system fields** as well which cannot be done using Data Import Wizard.

Some of the System Fields are :

- CreatedBy
- CreatedDate
- LastModifiedBy
- LastModifiedDate

To insert System Fields, we need to give one 'System Permission' to the User. We can do it by creating a 'Permission Set'.

Insert System Fields

To Create a Permission Set, follow this path:

Setup → Permission Set → New → System Permissions → Enable 'Set Audit Fields upon Record Creation' → Save. Assign the Permission Set to the User.



Using Dataloader.io to Insert Related Records Using External Id

What is External Id ?

A custom field of type Text, Email or Number can be configured as **External Id** which stores unique identifiers from external systems, such as an ERP or a database. It serves as a link between Salesforce records and the corresponding records in external systems.

In Dataloader.io, when importing related records, an external ID refers to a field in the related objects that uniquely identifies records and is used to establish relationships between them during the import process.



Export Data

Purpose of Data Export

Data Backup and Recovery: Regularly exporting data ensures that we have backups, essential for data recovery in case of data loss.

Data Migration: Exporting data is an important step when moving data from one system to another.

Data Analysis: We can analyze exported data for various purposes using external tools like MS Excel or Tableau.

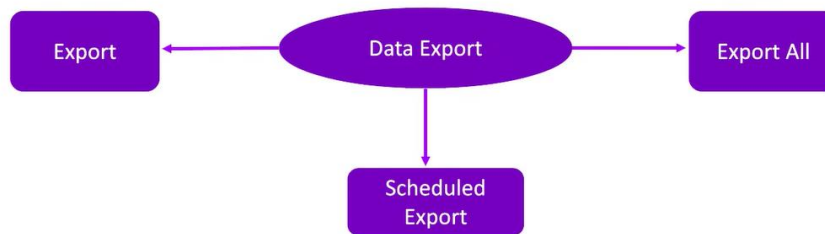
Data Sharing: We can share the exported data with our clients or other people who do not have the direct access to the system.

Data Transformation: Before importing data to other systems, we can transform the exported data according to the requirements while maintaining data integrity and quality.

Use Dataloader.io Export Data

Dataloader.io Export options

Data export refers to the process of extracting data from the Salesforce platform and transferring it to an external system. Data export is essential for various purposes such as data backup, analysis, and integration with other systems.



Dataloader.io Export options

- **Export:** This refers to manually extracting specific data from Salesforce Objects.
- **Export All:** The "Export All" feature in Salesforce allows administrators to export data from the Salesforce object, including records in the recycle bin.
- **Scheduled Export:** Scheduled exports in Salesforce enable users to automate the data extraction process on a recurring basis like Daily, Weekly, Monthly or Yearly.

Upsert Data

Upserting Data

Upsert combines the functionality of both Update and Insert into a single import operation. It allows us to either update an existing record or insert a new record depending on whether a match is found for a specified Record Id.

	A	B	C	D	E	F
Records with Record Id will get Updated	1 Record ID	Department	Duration	Education	Position Name	Priority
	2 a06IR00004qwWhmYAE	IT		Bachelor's degree	DBA	Medium
	3 a06IR00004qwWhnYAE	Engineering		High school diploma	Sr. UI Designer	Medium
	4	Engineering		Bachelor's Degree	UI Designer	High
	5 a06IR00004qwWhpYAE	IT		High school diploma	Sr. Director, Data Center	Critical
Records without Record Id will get inserted	6	Engineering		Bachelor's degree	SW Engineer	High
	7	Engineering		Bachelor's degree	Sr. SQA Engineer	Medium
	8 a06IR00004qwWhsYAE	Engineering		Bachelor's degree	Documentation Writer	High
	9 a06IR00004qwWhtYAE	IT		Bachelor's degree	Help Desk Analyst	Medium
	10	IT		Bachelor's degree	Network Technician	Critical
	11	IT		Bachelor's degree	Oracle Developer	High
	12					

Deleting Data

- 'Delete Data' refers to removing records from Salesforce database. When we delete the record, it moves to the recycle bin, where it can be restored.
- If we permanently delete the record from recycle bin, it is no longer recoverable.
- To delete data, we only need Record Id in the CSV file as displayed. We can delete the existing records upon matching the Record Id from the CSV file.

	A
1	Record ID
2	a06IR00004qwWhmYAE
3	a06IR00004qwWhnYAE
4	a06IR00004qwWhoYAE
5	a06IR00004qwWhpYAE
6	a06IR00004qwWhsYAE
7	a06IR00004qwWhtYAE
8	a06IR00004qwWhuYAE
9	a06IR00004qwWhvYAE
10	a06IR00004qwWhwYAE
11	a06IR00004qwWhxYAE

Data Import Wizard	Dataloader.io
1. Data Import Wizard can import up to 50000 records at a time.	1. Dataloader.io can import up to 1 million records at a time.
2. Data Import Wizard is an internal tool available with our Salesforce Org.	2. Dataloader.io is an external cloud-based application.
3. Data Import Wizard does not support Bulk Delete.	3. Dataloader.io supports Bulk Delete.
4. It supports importing standard objects, such as Accounts, Contacts, Leads, Solutions, Campaign members and custom objects.	4. It supports importing a wider range of standard and custom objects
5. It cannot import the system Fields like CreatedDate, CreatedByld etc.	5. It can import the system fields like CreatedDate, CreatedByld etc.
6. It does not offer automation features. Each import needs to be initiated manually through the wizard interface.	6. It provides scheduling capabilities, allowing users to schedule data loads at specified intervals.